

EXPERIMENT 5 – Inter Process Communication (IPC) using Pipe

Aim: To implement IPC between processes using the pipe mechanism in UNIX.

C Program:

```
#include<stdio.h>
#include<stdlib.h>
#include<unistd.h>
#include<string.h>
#include<sys/wait.h>

int main()
{
    int fd[2];
    pid_t pid;
    char message[] = "Hello from Child Process";
    char buffer[100];

    pipe(fd);

    pid = fork();

    if(pid == 0)
    {
        close(fd[0]);
        write(fd[1], message, strlen(message)+1);
        close(fd[1]);
        exit(0);
    }
    else
    {
        wait(NULL);
        close(fd[1]);
        read(fd[0], buffer, sizeof(buffer));
        printf("Message received from child: %s\n", buffer);
        close(fd[0]);
    }

    return 0;
}
```

Compilation: `gcc ipc\_pipe.c -o ipc\_pipe`

Output:

Message received from child: Hello from Child Process

Shell Script:

```
#!/bin/bash

(
echo "Message from Child Process"
) | while read msg
do
    echo "Parent Received: $msg"
done
```

Result: IPC using Pipe was implemented and verified successfully.
